Chapter 1

INTRODUCTION

1.1 Overview

Modern warfare has transitioned into an increasingly data-driven environment where split-second decisions are critical for survival and mission success. The sheer volume of visual information from surveillance feeds, drones, and tactical environments often overwhelms personnel, leading to cognitive fatigue and a higher margin for error. Manual threat assessment is no longer sufficient to ensure the safety and tactical superiority of ground units in high-stress scenarios. Consequently, there is an urgent need for intelligent, autonomous systems that can process vast amounts of real-time visual data to provide actionable insights and enhance situational awareness.

VeerDrishti is an AI-enabled combat vision system designed to bridge this gap by providing soldiers with a persistent, intelligent observer through a camera-integrated solution. By leveraging state-of-the-art computer vision models and edge computing, the system autonomously identifies enemy combatants, vehicles, and weapons. The primary objective is to offload the mental burden of constant monitoring from the soldier, allowing them to focus on strategic execution. Developed as a lightweight solution suitable for deployment on wearable hardware, the system aims to transform how visual data is utilized on the front lines of national security.

The project incorporates sophisticated methodologies to ensure reliability under diverse and challenging battlefield conditions. It utilizes multimodal image fusion—combining visible and thermal spectrums—to maintain high detection accuracy in low-light or obscured environments. Furthermore, by implementing edge-optimized architectures and advanced feature fusion techniques, the system achieves high-speed inference without compromising precision. This integration of versatile object detection and robust information processing ensures that the system remains a reliable asset, offering a significant technological advantage in modern combat and defense operations.

1.2 Problem Statement

The problem statement is to address the challenges of managing vast real-time visual data in modern battlefield environments, where manual monitoring is slow, error-prone, and inefficient; in order to enhance situational awareness and tactical decision-making, an AI-powered combat vision system is proposed to automatically detect, classify, and assess threats in real time, improving response speed and operational efficiency.

1.3 Motivation

The motivation for VeerDrishti is to minimize casualties and save lives by providing an intelligent, persistent observer in combat environments. Modern warfare involves overwhelming visual data that leads to cognitive fatigue, increasing the risk of fatal human error. The system serves as a technological safeguard, enhancing personnel safety and protecting national security forces during high-stress operations.

Scope: The scope of the project is to develop an autonomous, real-time vision system designed to detect enemy personnel, weapons, and vehicles in tactical environments. The technical implementation will focus on developing a lightweight system, optimized for localized inference on mobile edge devices.

Purpose: The purpose of the project is to provide an autonomous, intelligent observer that delivers real-time situational awareness in high-stress combat environments. By automating threat detection and mitigating cognitive fatigue, the system serves as a technological safeguard dedicated to achieving the project's core mission: minimizing casualties and saving lives.

Applicability: The project will be applicable in active combat operations, border surveillance, and high-risk tactical missions where ground units require instantaneous threat identification. The edge-optimized architecture will ensure reliability in signal-denied zones and low-visibility environments, serving as a critical force multiplier for national security personnel. Beyond defense, the system's ability to automate hazard detection will be directly transferable to emergency response and disaster management, fulfilling the mandate to minimize casualties and save lives.

1.4 Objectives

The project's primary objectives are to develop a lightweight, real-time combat vision system that can be deployed on wearable hardware, capable of autonomously detecting and classifying threats in dynamic battlefield environments.
Specific objectives for the project include:

i. Implement a lightweight real-time object detection model.

ii. Enable on-screen visual highlighting of identified threats.

iii. Optimize system performance for low latency and smooth operation.

iv. Evaluate accuracy and response time under simulated field conditions.

1.5 Methodology

The project adopts a hybrid deep-learning approach that integrates generative data augmentation with edge-optimized multimodal fusion. Initially, the system utilizes a Generative Adversarial Network (GAN) to synthesize high-quality military training data, addressing scarcity in combat-specific datasets. For real-time execution, the project implements a lightweight detection architecture, such as EfficientDet or YOLOv10, optimized for low-latency performance on edge devices like smartphones or Raspberry Pi. To ensure reliability in varying field conditions, the methodology incorporates multimodal image fusion, specifically merging RGB texture with Infrared (thermal) edge data using a fine-grained region attention mechanism. This ensures robust target classification even during night operations or in cluttered environments. Furthermore, the system leverages a vision-language model (like Florence v2) for zero-shot detection, allowing the identification of ”unseen” or novel threats through natural language supervision. Post-detection, the pipeline filters data and applies an Augmented Reality (AR) overlay to highlight threats instantly, while securing all mission metadata in a local storage layer for subsequent auditing.

1.6 Sustainable Development Goal Relevance

The project aligns with United Nations Sustainable Development Goal 16: Peace, Justice, and Strong Institutions.

Target 16.1: Significantly reduce all forms of violence and related death rates everywhere

VeerDrishti will enhance battlefield situational awareness by detecting and tracking potential threats in real time. This capability will enable defense personnel to respond faster and make more informed decisions during combat situations. By improving awareness and reaction time, the system will help reduce operational risks and potentially decrease casualties.

Target 16.3: Promote the rule of law at the national and international levels and ensure equal access to justice for all

VeerDrishti will record visual data during operations, enabling improved mission analysis and operational review. It will support evidence-based evaluation of decisions made during missions. Such recorded data could also assist investigations related to violations of international humanitarian law, including potential war crimes.

Target 16.A: Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime

VeerDrishti will integrate AI-based detection and tracking technologies, strengthening the technological capacity of national security institutions. The project will support peacekeeping operations and the protection of national interests.

1.7 Feasibility Study

The feasibility of the proposed combat vision system is assessed across technical, operational, and economic dimensions.

Technical Feasibility: The proposed system is technically feasible using existing smartphone hardware and computer vision frameworks. Modern phones provide high-resolution cameras, sufficient processing power, and GPU support for real-time object detection. Light weight AI models can be optimized for edge deployment, enabling on-device threat detection without requiring complex external hardware.

Operational Feasibility: The system can be integrated into existing helmet setups with minimal modification, using a mounted smartphone as both camera and display interface. It reduces cognitive load by automating threat detection and visual highlighting. Basic training would enable personnel to interpret AR overlays effectively in dynamic environments.

Economic Feasibility: Using commercially available smartphones significantly lowers development and deployment costs compared to specialized military hardware. Open-source AI frameworks further reduce software expenses. The prototype approach minimizes initial investment, making the system cost-effective for research, testing, and potential scalable improvements in future versions.

1.8 Work Distribution

Table 1.1: Work Distribution Among Team Members

Name	**Work**
Gaurav Chauhan	Literature Survey and Report Writing
Gaurav Singh	Literature Survey and Slide Preparation

Table 1.1 represents the work distribution between the team members for Major Project (Phase-1).

Chapter 2

LITERATURE SURVEY

2.1 Literature Review

X. Zhuang, et al. This paper presents an innovative approach to military target detection by combining EfficientDet with a Generative Adversarial Network to improve training data quality and detection accuracy. Initially, EfficientDet is applied to identify targets within the original dataset. Subsequently, the GAN generates synthetic military target images, which are incorporated into the training dataset to enhance its diversity and robustness. The expanded dataset is then used to retrain the EfficientDet model, leading to significantly improved detection performance, particularly in situations where limited training data is available. Experimental evaluation on a custom military target dataset demonstrated the effectiveness of the proposed method. The model achieved a mean Average Precision (mAP) of 92.8%, surpassing the performance of conventional approaches such as YOLOv3 and YOLOv5. Moreover, the system maintained real-time processing capability at 15 FPS and achieved over 90% detection accuracy under challenging conditions, including dim lighting and occlusions. **[1]**

C. Konar, et al. This paper presents a portable and edge-optimized surveillance system developed to detect modern security threats such as concealed weapons and drones. The proposed framework is designed for deployment in resource-constrained environments, including remote border regions and civilian infrastructure, while maintaining military-grade object detection capabilities. The system is built upon the Florence v2 Vision-Language Transformer and incorporates Parameter Efficient Fine-Tuning (PEFT) along with Low-Rank Adaptation to improve adaptation efficiency with reduced computational overhead. The model was trained using a dataset of 12,000 images, including GAN-generated synthetic samples to enhance detection robustness. For deployment, the framework utilized a Raspberry Pi 5 integrated with an ESP32-S2 module, enabling wireless alert transmission through the ESP-NOW communication protocol. Experimental results demonstrated the effectiveness of the proposed system, achieving an accuracy of 88.01% and outperforming YOLOv8 and Faster R-CNN. Additionally, the model attained a mAP_{50} score of 64.83 while maintaining high efficiency on edge hardware with low CPU and memory utilization. **[2]**

A. A. S. Matta , et al. This paper presents an optimized spatiotemporal deep learning framework for predictive behavioral threat detection in surveillance footage. The proposed system addresses challenges in anomaly detection, including complex human behavior, temporal variability, and limited annotated data, by integrating a Convolutional Neural Network with Long Short-Term Memory networks. The CNN component extracts spatial features from video frames, while the LSTM captures temporal dependencies across sequential frames, enabling effective recognition of suspicious activities. The framework also incorporates hyperparameter optimization, regularization techniques, and cross-validation strategies to improve model stability and generalization. Experimental evaluation was conducted using the DCSASS surveillance dataset containing multiple anomaly categories such as robbery, assault, and vandalism. Results demonstrated that the optimized CNN-LSTM model achieved an impressive accuracy of 98.1%, with high precision, recall, and F1-scores across 3-fold, 5-fold, and 10-fold validation settings. Furthermore, the proposed framework outperformed traditional machine learning models and recent deep learning approaches, proving highly effective for real-time intelligent surveillance applications. **[3]**

N. Li, et al. This paper proposes PDGV-DETR, an advanced framework for precise weapon and personnel detection in static security surveillance images. The study addresses critical challenges such as object occlusion and complex backgrounds, which often reduce the effectiveness of conventional detection models in public safety applications, including school and subway surveillance systems. The proposed framework enhances detection performance through three major components. First, a dynamic hierarchical channel interaction convolution module improves the recognition of partially occluded objects. Second, an improved bidirectional hybrid feature pyramid network enables effective cross-scale feature fusion for detecting objects of varying sizes. Finally, a global semantic weaving network strengthens the model's ability to distinguish targets from cluttered and complex backgrounds in static scenes. Experimental evaluation demonstrated the effectiveness of PDGV-DETR across multiple datasets. The framework achieved a peak mAP_{50} of 85.9% on conflict-scene datasets and 93% on weapon-specific datasets. Furthermore, the model significantly outperformed RT-DETR and Deformable DETR, while maintaining a balance between high localization accuracy and computational efficiency suitable for real-time surveillance applications. **[4]**

A. M. Garcia-Rubio, et al. This paper presents a collaborative AI-human framework for detecting concept drift in object detection models used in defense and surveillance applications. The proposed system addresses the degradation of object detection performance caused by evolving environments, lighting conditions, and unfamiliar objects. The framework integrates a YOLOv7 architecture with human feedback to improve reliability and adaptability in dynamic military scenarios. A custom metric called the Drift Avoidance Value was introduced to quantify concept drift by combining AI-generated confidence scores with human corrections, duplicate detection analysis, and missed object evaluations. The model was trained on 4,579 military vehicle images and evaluated using multiple datasets containing main battle tanks, humans, and unknown ground vehicles. Experimental results demonstrated that retraining the model with human-corrected data significantly improved detection performance. The Drift Avoidance Value increased after retraining, while prediction accuracy improved by 16.9% for tanks and 12.5% for humans. The study highlights the effectiveness of human-in-the-loop learning for maintaining robust and adaptive object detection systems in real-world defense environments. **[5]**

X. Xiang, et al. This paper introduces UniFusOD, an end-to-end framework designed for infrared–visible image fusion and object detection. The proposed system combines visible images, which provide rich texture and color information, with infrared images that offer strong edge details and reliable performance under low-light conditions. By integrating these complementary modalities, UniFusOD generates a single fused image optimized for both visual clarity and accurate object detection. The framework incorporates the FRA mechanism, which utilizes multiple convolution kernels to capture fine-grained details as well as broader semantic features. Additionally, the proposed UnityGrad Optimization strategy balances fusion and detection gradients, preventing conflicts between the two tasks and enabling more effective joint training. Experimental evaluation demonstrated that UniFusOD outperformed several state-of-the-art methods across multiple benchmark datasets. The model achieved improved mean Average Precision (mAP) scores, including gains of 0.8 on the DroneVehicle Dataset and 1.9 on the M3FD Dataset. Furthermore, the framework produced fused images with superior structural quality and information content while maintaining strong robustness in detecting small targets within complex environments. **[6]**

Z. Shou, et al. This paper presents a robust framework for cyber-physical systems, including autonomous vehicles and smart surveillance applications, through the development of MSAF-Net for human action recognition. The proposed system integrates multiple visual modalities such as RGB, depth, and thermal images to enhance detection accuracy and reliability in complex environments. The framework employs a Multi-Scale Attention Fusion mechanism that extracts features at different levels of detail while emphasizing the most relevant information for accurate recognition. In addition, Cross-Level Feature Interaction enables different network layers to exchange information, preserving both fine-grained details and global contextual structures. The model also incorporates an Adaptive Fusion Strategy that combines semantic and structural guidance to generate clear and consistent fused representations. Experimental evaluation on several benchmark datasets demonstrated the effectiveness of the proposed approach. MSAF-Net achieved 91.54% accuracy on the UCF101 Dataset and 92.14% accuracy on the ActivityNet, outperforming existing state-of-the-art methods. Furthermore, the framework remained robust even when individual sensors failed and could be efficiently pruned for deployment on low-power hardware platforms. **[7]**

Z. Song, et al. This paper introduces EDNet, a deep learning framework designed for detecting small objects in UAV and drone imagery. Built upon YOLOv10, the framework is optimized for edge computing, enabling efficient deployment on mobile and low-power devices. EDNet is highly scalable and offers seven model variants ranging from Tiny, which is lightweight and computationally efficient, to XL, which provides high-performance detection capabilities. The architecture incorporates the C2f-FCA block, which utilizes faster context attention mechanisms to extract meaningful features while maintaining low computational complexity. In addition, the model employs the WIoUv3 Loss Function to prioritize high-quality samples and reduce the influence of noisy data during training. For optimized deployment, the framework supports technologies such as CoreML, TensorRT, and OpenVINO, ensuring high-speed performance across multiple hardware platforms. Experimental results demonstrated that EDNet outperformed YOLOv8 and YOLOv10 in both accuracy and efficiency, achieving up to 5.6% higher mAP_{50} with fewer parameters while maintaining real-time performance across devices such as the iPhone 12 and Raspberry Pi 5. **[8]**

S. D N, et al. This paper presents a state-of-the-art object detection system designed for analyzing camera and drone footage in military environments. The proposed framework focuses on identifying battlefield threats such as vehicles, weapons, and combatants in real time. The system employs a vision-language model that combines image data with natural language descriptions to improve object understanding and classification accuracy. It integrates CLIP to associate images with textual descriptions, enabling the detection of unseen objects without requiring extensive labeled datasets. Additionally, the SPPELAN pooling module is used to generate fixed feature representations while minimizing information loss. The framework also incorporates an N-gram and GLIP labeler to extract noun phrases and create pseudo-boundaries for automated training data generation. Furthermore, K-means Clustering is applied to group similar image regions, improving detection efficiency in cluttered environments. Experimental evaluation on the Microsoft COCO Dataset demonstrated superior performance compared to Grounding DINO, achieving lower complexity, higher accuracy, and significantly faster frame rates suitable for real-time deployment. **[9]**

V. Kaya, et al. This paper proposes a deep learning–based approach for the automatic detection and classification of seven different handheld weapon types. The study aims to assist security agencies by developing a proactive surveillance system capable of identifying threatening objects in real time, thereby helping to prevent criminal activities. The proposed model is based on the VGGNet architecture and consists of 25 layers with 337,671 parameters, making it both efficient and computationally lightweight. A custom dataset containing 5,214 images of various weapons, including assault rifles, pistols, and grenades, was collected from internet sources and preprocessed using grayscale conversion and background cleaning techniques. The model was implemented and trained using the Keras deep learning framework. Experimental results demonstrated that the proposed CNN model achieved an impressive classification accuracy of 98.40%, significantly outperforming models such as VGG-16, ResNet50, and ResNet-101. Additionally, the model achieved a low loss rate of

0.052, highlighting its reliability, fast training capability, and suitability for deployment on low-cost computing systems. **[10]**

Ankita, S. Rani, et al. This paper introduces an efficient and lightweight deep learning framework for human activity recognition using smartphone sensor data. The study addresses the limitations of traditional pattern recognition techniques by employing automated feature extraction, which is highly beneficial for applications such as healthcare, elderly assistance, and sports science. The proposed model integrates a Convolutional Neural Network with Long Short-Term Memory units to form a CNN-LSTM architecture capable of learning both spatial and temporal features. Convolutional layers automatically extract meaningful patterns from accelerometer and gyroscope signals, while LSTM layers analyze sequential dependencies in human activities. The model was trained and evaluated using the UCI-HAR Dataset collected from Samsung Galaxy S2 smartphones, with Gaussian standardization applied during preprocessing. Experimental results demonstrated that the CNN-LSTM model achieved an impressive recognition accuracy of 97.89%. Furthermore, the model outperformed traditional machine learning approaches by effectively recognizing activities such as walking, sitting, and standing while maintaining a lightweight structure suitable for mobile deployment. **[11]**

2.2 Comparative Study Of References

Table 2.1: Comparison Table of Research Papers

Sl. No.	Research Paper	Learning Outcome
1	**Title:** Military target detection method based on EfficientDet and Generative Adversarial Network **Authors:** X. Zhuang, D. Li, Y. Wang, and K. Li **Published:** Engineering Applications of Artificial Intelligence, vol. 132, Jun. 2024	**Objective:** To detect targets in low visibility, noisy, or data-scarce combat zones **Methodology:** EfficientDet-D0 + Dual-Discriminator GAN **Core Idea:** Uses GANs to generate synthetic military training data and BiFPN for feature fusion

Sl. No.	Research Paper	Learning Outcome
2	**Title:** Edge-Optimized Threat Detection with Florence v2 **Authors:** C. Konar, D. Das, D. S., K. Jc, and S. Singh **Published:** PeerJ Computer Science, vol. 12, Mar. 2026	**Objective:** To create a portable, edgeoptimized surveillance system capable of detecting military-relevant threat classes in resourceconstrained environments **Methodology:** Florence v2 Vision Language Transformer **Core Idea:** Leveraging multimodal vision-language models on compact edge hardware to provide situational awareness without relying on cloud infrastructure
3	**Title:** Spatiotemporal deep learning framework for predictive behavioral threat detection in surveillance footage **Authors:** A. A. S. Matta and V. P. C. S. R. Manukonda **Published:** Frontiers in Big Data, vol. 9, Feb 2026.	**Objective:** To detect anomalous human behaviors in untrimmed surveillance footage using optimized spatiotemporal deep learning **Methodology:** Hybrid CNN-LSTM framework on DCSASS dataset **Core Idea:** Integrating CNN spatial features and LSTM temporal patterns to reliably identify 13 distinct types of criminal behavior
4	**Title:** PDGV-DETR: Object Detection for Secure On-Site Weapon and Personnel Location **Authors:** N. Li et al. **Published:** Sensors, vol. 26, Feb. 2026	**Objective:** To precisely detect and position weapons and personnel in static surveillance images while overcoming occlusion and complex background interference **Methodology:** PDGV-DETR framework **Core Idea:** Transformer based end-to-end detection system that uses cross-scale semantic fusion to bridge the gap between high-level semantics and low-level image details

Sl. No.	Research Paper	Learning Outcome
5	**Title:** Detecting Concept Drift in Object Detection Models: A Collaborative AI-Human Approach for Defense Applications **Authors:** A. M. Garcia-Rubio, L. Heck, K. M. Schweitzer, R. M. Bateman, A. Alaeddini, and K. K. Castillo-Villar **Published:** IEEE Access, vol. 14, Jan. 2026	**Objective:** To quantify and manage model performance deterioration in military object detection systems over time **Methodology:** YOLO confidence scores + direct human feedback. **Core Idea:** Enhancing AI reliability by integrating human corrections directly into drift calculations to guide targeted and efficient model retraining
6	**Title:** Infrared-Visible Image Fusion Meets Object Detection **Authors:** X. Xiang et al. **Published:** Remote Sensing, vol. 17, Nov. 2025	**Objective:** To solve gradient conflicts when training fusion and detection together **Methodology:** UniFusOD **Core Idea:** Use Fine-Grained Region Attention (FRA) mechanism and unified framework to fuse RGB and infrared images for accurate detection
7	**Title:** Multi-modal action recognition via advanced image fusion techniques **Authors:** Z. Shou and D. Zhu **Published:** Frontiers in Physics, Aug. 2025	**Objective:** To recognize human/combatant actions accurately even if one sensor fails **Methodology:** MSAF-Net (Multi-Scale Attention-Guided Fusion) **Core Idea:** Dynamically prioritizes sensor data and switches to backup sensors when needed

Sl. No.	Research Paper	Learning Outcome
8	**Title:** EDNet: Edge-Optimized Small Target Detection in UAV Imagery **Authors:** Z. Song, Y. Zhang, and A. A. R. M. A. Ebayyeh **Published:** IEEE SWC, Dec. 2024	**Objective:** To enable high-speed detection of tiny objects on low-power drone hardware **Methodology:** YOLOv10 + Faster Context Attention (C2fFCA) **Core Idea:** Uses ”XSmall” detection layer for small object detection with hardware acceleration optimization
9	**Title:** Real-Time Versatile Object Detection With High Accuracy And Reduced Information Loss During Wartime **Authors:** S. D. N. and D. M. **Published:** IEEE Conference, Sep. 2024	**Objective:** To detect unseen or new battlefield threats without manual re-labeling **Methodology:** CLIP Encoder + Modified C2f-Darknet **Core Idea:** Uses pseudo-boundary labeling for zero-shot detection and combines vision and language
10	**Title:** Detection and Classification of Different Weapon Types Using Deep Learning **Authors:** V. Kaya, S. Tuncer, and A. Baran **Published:** Applied Sciences, vol. 11, Aug. 2021	**Objective:** To propose a model for the automatic detection and classification of distinct types of handheld weapons **Methodology:** CNN based on VGGNet architecture **Core Idea:** Utilizing reduced layer complexity and highprecision CNNs to enable weapon identification on low-cost computing systems

Sl. No.	Research Paper	Learning Outcome
11	**Title:** An Efficient and Lightweight Deep Learning Model for Human Activity Recognition Using Smartphones **Authors:** Ankita, S. Rani, H. Babbar, S. Coleman, A. Singh, and H. M. Aljahdali **Published:** Sensors, vol. 21, Jun. 2021	**Objective:** To develop an efficient and lightweight model for identifying human activities using smartphone-based sensor data **Methodology:** CNN + Long Short-Term Memory (LSTM) **Core Idea:** Integrating spatial and temporal deep learning architectures to automate feature extraction from raw sensor data

Table 2.1 represents the comparison between the different research papers used for the literature review.

Chapter 3

REQUIREMENTS AND ANALYSIS

The requirements analysis for the VeerDrishti combat vision system encompasses a comprehensive evaluation of functional and non-functional requirements, ensuring that the system meets the operational needs of modern warfare while adhering to technical constraints. The analysis is structured to identify key features, performance metrics, and user interactions necessary for the successful deployment of the system in dynamic battlefield environments.

3.1 Functional Requirements

The functional requirements define the core capabilities of the project, including:

i. **Real-Time Object Detection:** The system must be capable of detecting and classifying combatants, vehicles, and weapons in real time using the integrated camera feed.

ii. **Visual Highlighting:** Detected threats should be visually highlighted on the display interface, providing clear and immediate feedback to the user.

iii. **Open Vocabulary Recognition:** The system should be able to identify novel or previously unseen threats using natural language supervision and language-vision encoders.

iv. **User Interface:** The system must provide an intuitive user interface that allows easy interpretation of the visual information to make informed decisions without distraction.

v. **Data Logging:** The system must provide secure local storage of detection events and video metadata for post-mission analysis and system auditing.

3.2 Non-Functional Requirements

The non-functional requirements address the performance, reliability, and usability aspects of the project, including:

i. **Low Latency:** The system must operate with minimal latency to ensure timely threat detection and response in high-stress combat scenarios.

ii. **Robustness:** The system should maintain high detection accuracy under varying environmental conditions, such as low light, occlusion, and dynamic backgrounds.

iii. **Scalability:** The system should be designed to accommodate future enhancements, such as additional threat categories or integration with other battlefield systems.

iv. **Security:** The system must ensure the confidentiality and integrity of the data collected, preventing unauthorized access and ensuring compliance with military data handling standards.

v. **Usability:** The system should be user-friendly, requiring minimal training for effective operation, and should not contribute to cognitive overload for the user.

3.3 Hardware Requirements

i. **Development Workstation:** High-performance laptop equipped with a dedicated GPU for code development, dataset preprocessing, and initial model debugging.

ii. **Edge Deployment Device:** High-end smartphone with an integrated Neural Processing Unit (NPU) to host the prototype application and execute real-time local inference.

iii. **Cloud Training Infrastructure:** Scalable cloud-based GPU instances utilized for training complex deep learning architectures and performing large-scale hyperparameter optimization.

3.4 Software Requirements

i. **Operating System:** Linux distribution like Ubuntu 26.04 LTS with stable CUDA/ROS support.

ii. **Programming Languages:** Python 3.9+ for model development, Rust for high-performance edge execution and Kotlin for mobile application development.

iii. **IDE/Tools:** VS Code, Jupyter Notebooks for experimentation, and Git for version control.

iv. **Core Framework:** PyTorch 2.0+ or TensorFlow 2.x (primarily PyTorch for YOLOv8/v10 implementation).

v. **Cloud Platform:** Google Cloud Vertex AI or AWS EC2 for large-scale model training.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

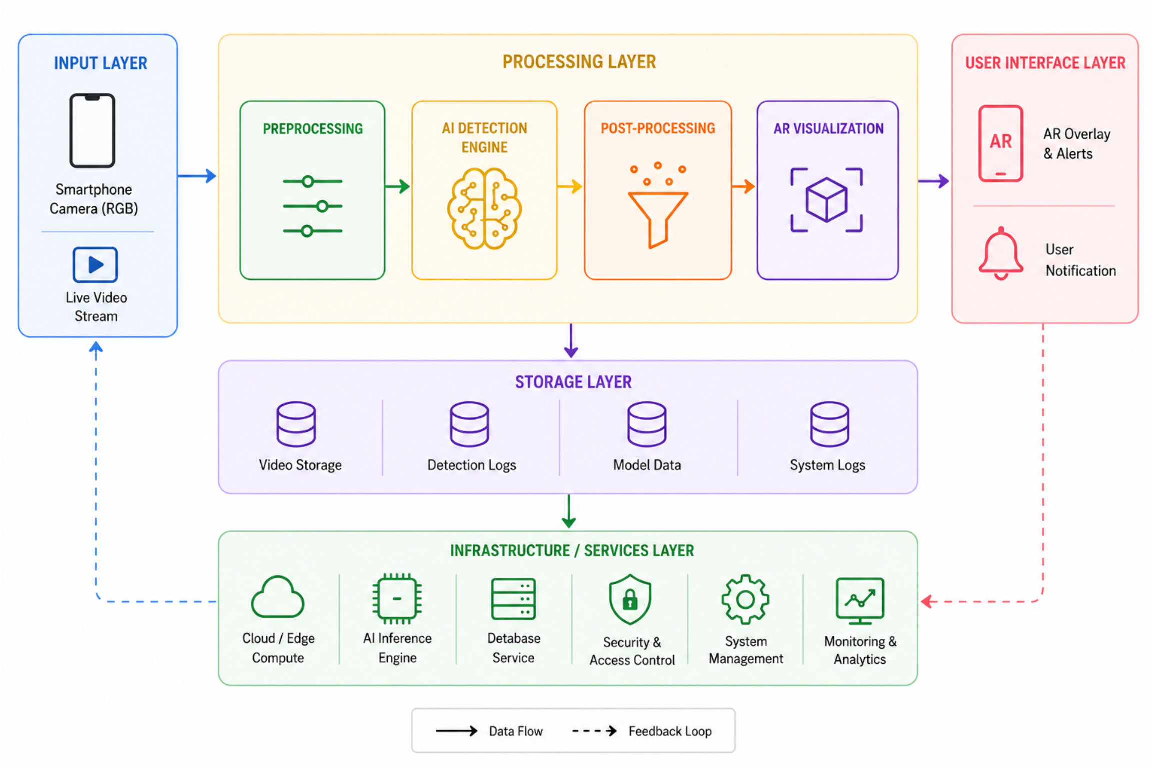

Fig. 4.1: System Architecture

Fig. 4.1 represents the tentative system architecture of the VeerDrishti project. The system architecture designed for the project is organized into six distinct and interconnected layers that collectively facilitate a smooth and continuous flow of data from the initial capture stage to the final storage stage. The architecture begins with the Input Layer, which is responsible for capturing real-time video footage and supplying the raw visual data required for further analysis and processing. This raw input data is then transferred to the Preprocessing Layer, where several enhancement operations are performed, including image resizing, alignment, and noise reduction, in order to improve image clarity and ensure that the data is suitable for accurate analysis.

The Processing Layer acts as the core intelligence hub of the entire system architecture. In this layer, objects are detected and specific scenarios are recognized using the processed visual information. Once the detections are completed, the data moves to the Post-Processing Layer, where the identified detections are filtered and further classified to determine potential threats and improve the reliability of the results.

Finally, the User Interface Layer provides an interactive user experience by displaying an AR overlay containing the processed results and detections in a visually accessible manner. Alongside this, the Storage Layer maintains logs of all detection data and related metadata, ensuring that the information is preserved for future reference, analysis, and system evaluation.

4.2 System Workflow

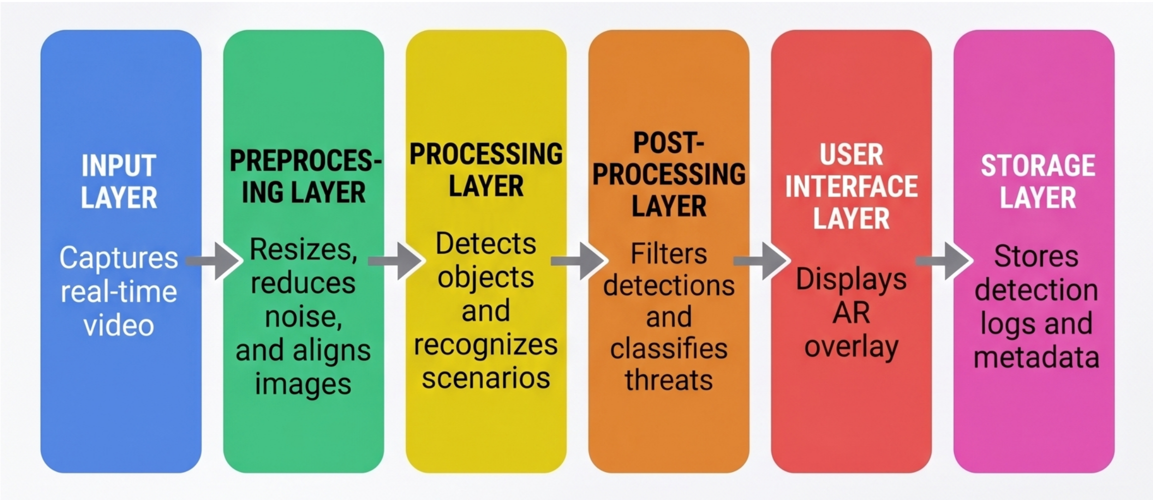

Fig. 4.2: System Workflow

Fig. 4.2 represents the tentative system workflow of the VeerDrishti project. The workflow is designed as a continuous and systematic loop that supports real-time monitoring and enables rapid response during operation. Upon initiation, the system begins by capturing live RGB video footage, which serves as the primary source of input for further processing and analysis. Immediately after capture, the video feed undergoes preprocessing operations such as denoising and normalization in order to improve image quality and maintain consistency for accurate detection.

Once preprocessing is completed, the AI Detection Engine analyzes the video feed to identify combatants, weapons, or vehicles present within the monitored environment. This stage acts as the core analytical component of the workflow, enabling the system to process visual data continuously and identify relevant objects or scenarios in real time. The workflow then reaches a critical decision point where the system determines whether a threat has been detected or not.

If a threat is identified, the system responds by applying bounding boxes around the detected objects, highlighting the danger through AR visualization, alerting the user, and logging the corresponding detection data for future reference. On the other hand, if no threat is detected, the system continues its monitoring cycle without interruption and returns to the capture stage.

Chapter 5

SCHEDULING

5.1 GANTT Chart

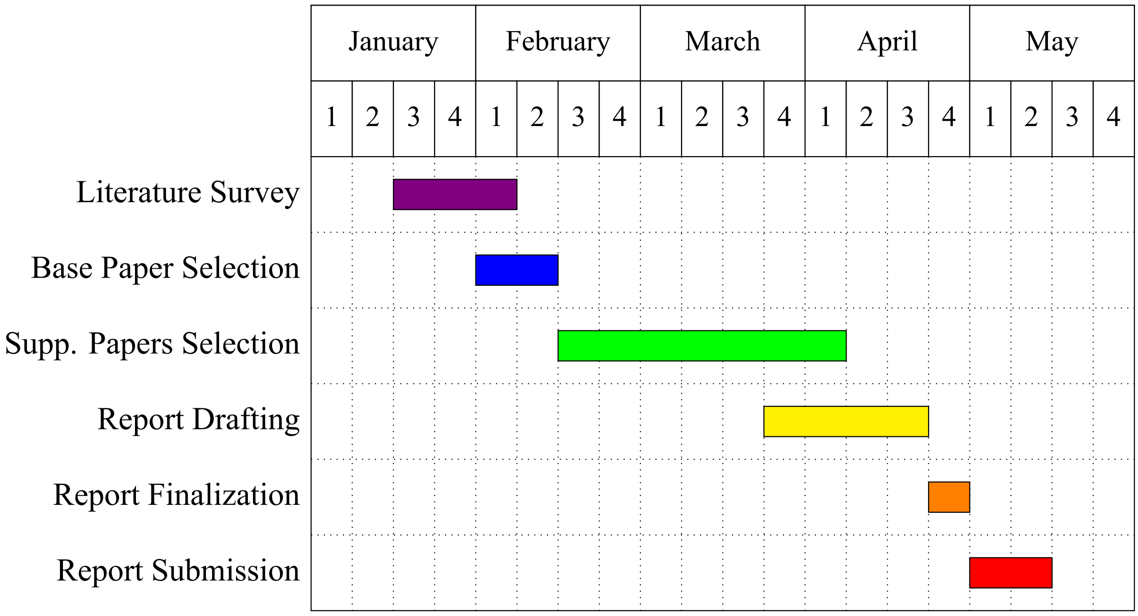

Fig. 5.1: GANTT Chart for Major Project (Phase-1)

Figure 5.1 represents the timeline for the Major Project (Phase-1). The project follows a well-structured and systematic timeline spanning from January to May, ensuring that each stage is completed in a sequential and organized manner. The initial phase of the project begins in the second week of January with a three-week Literature Survey, which continues steadily and concludes in early February. This stage establishes the research background and provides the necessary understanding for the subsequent activities. Immediately after this, the Base Paper Selection phase is carried out during the first two weeks of February, allowing the project to identify the primary reference work for further study and implementation. Supporting Papers Selection forms the longest and most extensive research block in the schedule. This phase begins in late February and extends continuously through the end of April, reflecting the importance of gathering and analyzing additional research materials relevant to the project topic. As the research work approaches completion, the Report Drafting phase starts in early April and continues for three weeks, focusing on organizing and documenting the findings systematically. The final stages are closely scheduled in May. Report Finalization takes place during the first week, followed by the final Report Submission in the second week of the month. This sequential and carefully planned approach ensures a rigorous research foundation before the final documentation is completed and formally submitted.

5.2 Month-wise Progress

The project timeline spans from January to May 2026, transitioning from initial literature research to final documentation. The month-wise progress details are as follows:

Table 5.1: Comparison Table of Research Papers

Sl. No.	Month	Progress Details
1	January	Conducted a comprehensive literature survey to identify relevant research papers and technologies in the domain of real-time combat vision systems. Identified research papers that could serve as the base paper and foundation for our project.
2	February	Selected the base paper and conducted extensive reading and analysis to understand the current state of the art, identify gaps in existing solutions, and select a base paper that aligns with our project objectives.
3	March	Selected 10 supplementary papers that provide additional insights, methodologies, and technologies relevant to our project. These papers contribute to the development of our combat vision system by providing a broader understanding of the field and potential approaches to implementation. Began intial work to draft the project report.
4	April	Drafted the project report, synthesizing information from the base paper and supplementary papers. Structured the report to clearly articulate the project details and specifications. Iteratively refined the report based on feedback and additional insights gained during the drafting process and drafted the final project report.
5	May	Submitted the finalized project report, ensuring that all sections are complete, well-organized, and adhere to the required guidelines.

Table 5.1 represents the month-wise progress of the Major Project (Phase-1).

Chapter 6

CONCLUSION

The project advances tactical situational awareness by combining state-of-the-art AI with wearable combat technology. It employs the lightweight YOLOv8 architecture and multimodal image fusion to enable real-time, autonomous threat detection in high-stress environments. By integrating visible and thermal imaging, the system remains effective in challenging conditions such as darkness, smoke, and low visibility. This reduces cognitive load on personnel by converting complex visual data into actionable insights, enabling faster and more accurate decision-making. The system's robustness is enhanced through GAN-based synthetic data augmentation, which addresses limited military datasets, and CLIP-based open-vocabulary detection, allowing adaptation to unseen threats via natural language guidance. Additionally, its edge-optimized design enables high-speed, low-latency inference on portable, smartphone-based hardware. This ensures reliable performance even in signal-denied environments while maintaining data security and operational efficiency, ultimately supporting rapid and informed responses in critical combat scenarios.

Looking forward, the potential for VeerDrishti extends into a scalable framework for future defense innovations. Future iterations could integrate more complex action recognition models to predict hostile intent or incorporate advanced feature-fusion strategies to further enhance the detection of micro-scale targets. In conclusion, VeerDrishti offers a versatile and reliable solution for modern national security. By harmonizing cutting-edge computer vision research with the practical, high-stakes requirements of the frontline, the project provides a definitive technological advantage, ensuring enhanced safety and operational efficiency for ground units in the evolving landscape of modern warfare.

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